Oncology, DSD, Gender & Ageing

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · **P2** = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 24	Basic Principles of Immunology and Immunotherapy in Urologic Oncology
Ch 25	Molecular Genetics and Cancer Biology
Ch 55	Management of Abnormalities of the External Genitalia in Males
Ch 58	Management of Abnormalities of the Genitalia in Girls
Ch 59	Differences of Sexual Development
Ch 60	Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations
Ch 65	Adolescent and Transitional Urology
Ch 120	Aging and Geriatric Urology
Ch 126	Management and Considerations in Transgender and Nonbinary Individuals
Ch 127	Male-to-Female Gender-Affirming Surgery: Vaginoplasty
Ch 128	Special Urologic Considerations in Transmasculine Individuals
★	Putting it together

24 Basic Principles of Immunology and Immunotherapy in Urologic Oncology

1 · INNATE AND ADAPTIVE IMMUNITY

P2 *Describe use and principles of immunology and Immunotherapy in Urological Oncology — innate/adaptive arms, TLR/PAMP, two signals, Th1 vs Th2, checkpoints, adaptive immune resistance, three Es*

Not in the paper answer:

- Th1 cytokines (IL-2, TNF-α, IFN-γ) mean **BCG success**; Th2 cytokines (IL-4, IL-5, IL-10, IL-13) mean **BCG failure**.

ALGORITHM

1. Danger signal (TNF-α, TLR ligand) reaches a DC.
2. DC stops antigen uptake; raises MHC and B7-1/B7-2.
3. DC enters afferent lymphatics (CCL21 sensed by CCR7).
4. In the node it activates CD4, CD8 and B cells.

	Class I MHC	Class II MHC
Recognised by	CD8	CD4
Distribution	Almost all cells; raised by IFN	Antigen-presenting cells (APCs)
Peptide	~9 amino acids	13–25 amino acids

2 · IMMUNE EDITING AND ESCAPE

- Escape follows equilibrium, as antigen and MHC fall and Tregs rise.
- In adaptive immune resistance, CD8-derived IFN-γ induces tumour PD-L1, which switches off those PD-1-positive CD8 cells.

KEY POINT

PD-L1 is mostly induced by the immune attack, so it marks a suppressed response, not a fixed trait.

T-cell molecule	Ligand	Effect
CD28	CD80/CD86 (B7)	Activating
CTLA-4	CD80/CD86	Inhibitory; mainly on FoxP3+ Tregs
PD-1	PD-L1 (B7-H1), PD-L2	Inhibitory; mainly on CD8 tumour-infiltrating lymphocytes (TILs)
LAG-3	MHC class II	Inhibitory

3 · INFLAMMATION AND GENITOURINARY (GU) TUMOURS

	Bladder	Kidney	Prostate
Does inflammation cause cancer?	Strong: schistosomiasis, chronic UTI, catheters, cyclophosphamide	Weak	Probable: proliferative inflammatory atrophy beside HGPIN
CD8 infiltration	Better outcome in MIBC	**Worse outcome — renal paradox**	May be suppressive
Dominant restraint	PD-L1, rising with grade	PD-1/LAG-3-high T cells, few APCs	Tregs; PD-L1-negative tumour

4 · BCG IN BLADDER CANCER

ALGORITHM

1. Instil 1–5 × 10^8 bacilli.
2. A fraction adheres via **fibronectin attachment protein**.
3. Urothelial TLR2/TLR4 release IL-8 and TNF-α.
4. Neutrophil wave (hours), then DC activation.
5. CD4 Th1 response in nodes (weeks), then memory.

- Memory explains durable remission in ~**50%** of NMIBC.
- **Nadofaragene firadenovec**, an adenovirus carrying the IFN-α2b gene with **Syn3**, is given 3-monthly for up to 12 months (CIS CR 53.4%). It was the FDA's **first gene therapy** (2022), for BCG-unresponsive NMIBC with CIS.

5 · CANCER VACCINES

P4 *Rationale of using cancer vaccines in Urology — antigen + GM-CSF adjuvant; sipuleucel-T (PAP–GM-CSF, 3 doses 2-weekly, IMPACT); failed IMA901, AGS-003, PROSTVAC*

- Sipuleucel-T, the first FDA-approved solid-tumour vaccine, prolongs OS by **+4.1 months** with **no effect on time to progression**.

6 · CHECKPOINT BLOCKADE

Agent	Target
Nivolumab, pembrolizumab	PD-1
Avelumab, atezolizumab, durvalumab	PD-L1
Ipilimumab	CTLA-4

Bladder

- First-line pembrolizumab alone (KEYNOTE-052) requires cisplatin-ineligibility with CPS **≥10**, or platinum-ineligibility; with CPS <10 it was worse than carboplatin.
- Atezolizumab and durvalumab were withdrawn, and adjuvant atezolizumab (IMvigor010) was negative.
- **Enfortumab vedotin (EV)**, an anti-nectin-4 antibody–drug conjugate, gave a 68% response rate with pembrolizumab when cisplatin-ineligible (EV-103).
- †In EV-302, EV plus pembrolizumab roughly doubled PFS and OS versus platinum and is now first line for all.

Kidney

- CTLA-4 on Tregs and PD-1 on CD8 TILs make dual blockade synergistic.
- VEGF blocks DC maturation and T-cell entry, so TKIs complement immunotherapy.

Trial (vs sunitinib)	Combination	ORR	Median PFS, months
KEYNOTE-426	Pembrolizumab + axitinib	60% vs 40%	15.4 vs 11.1
JAVELIN Renal 101	Avelumab + axitinib (only anti-PD-L1)	52.5% vs 27.3%	13.3 vs 8.0
CLEAR	Pembrolizumab + lenvatinib (best)	71% vs 36%	23.9 vs 9.2
CheckMate 9ER	Nivolumab + cabozantinib	55.7% vs 27.1%	16.6 vs 8.3

Prostate

- In mCRPC, tumours are rarely PD-L1-positive; pembrolizumab gives a PSA response under 10%, and ipilimumab and IMbassador250 gave no OS benefit.
- The exception is **MSI-H/dMMR** tumours (1.2–12%), treated with pembrolizumab second line or later; its 2017 approval was the **first based on a biomarker, not a site**.
- †Immune-related adverse events of grade ≥2 (such as colitis or pneumonitis) need drug hold and corticosteroids; endocrinopathies need lifelong replacement.

> **KEY POINT**
> Successful GU immunotherapy either supplies a danger signal (BCG, adjuvant, VEGF inhibition) or removes a brake (PD-1, CTLA-4); it fails where the brake is not limiting, as in Treg-dominated prostate cancer.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
†IMPACT	Sipuleucel-T vs placebo (2:1), mCRPC	OS 25.8 vs 21.7 months	Survival without progression delay
KEYNOTE-057	Pembrolizumab, BCG-unresponsive CIS	CR 41% at 3 months	FDA 2020
KEYNOTE-045	Pembrolizumab vs chemotherapy, post-platinum	OS 10.3 vs 7.4 months	Only phase III-based second-line approval
JAVELIN Bladder 100	Avelumab maintenance after platinum	Longer OS	Standard switch maintenance
CheckMate 274	Adjuvant nivolumab vs placebo, high-risk MIBC	DFS 20.8 vs 10.8 months	Adjuvant option
CheckMate 025	Nivolumab vs everolimus, RCC after TKI	Longer OS, less toxicity	First RCC checkpoint approval
CheckMate 214	Nivolumab + ipilimumab vs sunitinib, first line	OS hazard ratio 0.63	First line intermediate/poor risk
KEYNOTE-564	Adjuvant pembrolizumab for 1 year, high-risk clear-cell RCC	Recurrence or death reduced by 32%	Adjuvant standard

25 Molecular Genetics and Cancer Biology

1 · HALLMARKS AND BASIC MOLECULAR GENETICS

- **Local invasion** is the one hallmark that uniquely separates malignant from benign.
- **Alternative splicing** makes ligand-independent AR variants (AR-V7), causing castration resistance.
- **Somatic** testing finds targetable mutations; **germline** testing finds hereditary syndromes for surveillance and family cascade testing.

2 · GENE ALTERATIONS: TUMOUR SUPPRESSORS AND ONCOGENES

Oncogenes need one altered allele, suppressors usually two.

P1 *Oncogenes — gain of function/one allele vs suppressor loss/both alleles; mutation, amplification, fusion (TMPRSS2–ERG ~half, MYC, MET); Knudson two-hit*

Not in the paper answer:

- Two-hit exceptions are **dominant negative** mutation (mutant TP53) and **haploinsufficiency** (p27/CDKN1B).

3 · CELL CYCLE

- The cycle takes ~24 h, and cyclin-dependent kinases (CDKs) work only with their phase-specific **cyclin**.

Phase	Complex	Function
G1	Cyclin D–CDK4/6, cyclin E–CDK2	Phosphorylate RB1; cyclin D needs sustained mitogen (Ras/Raf/MEK)
S	Cyclin A–CDK2, then cyclin A–CDK1	S-phase genes
G2/M	Cyclin B–CDK1	Phosphorylates lamins, so the nuclear envelope breaks down

Checkpoint	Trigger	Mechanism
G1/S	Mitogens, nutrients, O_2, DNA damage	p21 (CDKN1A) and 14-3-3σ inactivate G1 CDKs; TP53 triggers apoptosis if damage is unrepaired
G2/M	DNA damage (radiation)	ATM activates CHEK1/CHEK2, which sequester Cdc25; TP53 induces p21 and GADD45
Spindle assembly	Unattached chromosome	Blocks anaphase-promoting complex; failure causes non-disjunction and **aneuploidy** (near-universal in solid tumours)

- CDK inhibitors are **Cip/Kip** (p21, p27, p57), acting broadly, and **INK4** (p15, p16, p18, p19), acting on CDK4/6.

RB1 — guardian of G1/S

ALGORITHM

1. Hypophosphorylated RB1 binds E2F and HDACs, silencing S-phase genes.
2. Cyclin D–CDK4/6 and cyclin E–CDK2 hyperphosphorylate RB1.
3. E2F is released and the restriction point is passed (irreversible).

- **TP53**, activated by DNA damage or oncogene stress, causes arrest, repair, senescence or apoptosis. **MDM2** is its negative regulator; TP53 is mutated in over half of cancers.

KEY POINT

Cancer must break both RB1 (entry) and TP53 (integrity), so both pathways are lost in almost every tumour.

4 · DNA DAMAGE AND REPAIR

Pathway	Lesion	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky adducts, cross-links (UV, chemicals)	XPC, XPD, ERCC1/XPF	Xeroderma pigmentosum; **ERCC2 in bladder cancer**
Base excision repair (BER)	Single-base damage, single-strand breaks	MUTYH, OGG1, PARP1/2	OGG1 variants linked to RCC
Mismatch repair (MMR)	Mispairs, indel loops at repeats	MSH2/MSH6, MLH1/PMS2	Mutation rate raised ~100-fold; Lynch
Homologous recombination (HR)	Double-strand break (DSB)	ATM, RAD51, BRCA1, BRCA2	Breast/ovarian, prostate
Non-homologous end joining (NHEJ)	DSB	Ku70/Ku80, DNA-PKcs, ligase IV	Translocations

- Unrepaired DSBs (marked by **γH2AX** foci) cause apoptosis, the basis of radiotherapy. **HR** copies the sister chromatid and is precise; **NHEJ** is template-free and error-prone.

KEY POINT

Repair defects are targets: HR loss means PARP inhibitor sensitivity, MMR loss means immunotherapy, and ERCC2 loss means cisplatin sensitivity.

5 · APOPTOSIS

- In apoptosis, initiator caspases 8, 9, 10 activate executioners 3, 6, 7.

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Unrepaired DNA damage, lost survival signals	Fas ligand/TNF on Fas (CD95)/TNFR1
Sequence	Bax/Bak release cytochrome c; Apaf-1 apoptosome activates caspase-9	FADD recruits procaspase-8, giving caspase-8
Brakes	Bcl-2, Bcl-XL, Mcl-1; IAPs	FLIP
TP53	Dependent (transactivates Bax)	**Independent**, so a target when TP53 is mutated

- Bid links the two pathways. TP53 loss causes chemoradiotherapy resistance.

6 · SENESCENCE, TELOMERES, AUTOPHAGY

P3 *Telomerase — ~100 bp lost/division, two barriers, then telomerase immortalisation; normal tissue expression, ALT (ATRX/DAXX); short telomeres in premalignant lesions, predict prostate recurrence*

Not in the paper answer:

- Telomerase is high in **85–90%** of tumours.
- **Senescent** cells are viable but permanently arrested; ADT induces senescence in some prostate cancers.
- **Autophagy** suppresses early tumours but sustains established ones.

7 · EPIGENETICS

- Epigenetic change alters expression without sequence change and is **reversible**. DNA methylation at CpG islands is maintained by DNMT1.

- Promoter hypermethylation silences suppressors as a "second hit": **GSTP1 in >90% of prostate cancer** (commonest somatic change), RASSF1A (up to 97% of bladder cancer) and p16. Global hypomethylation causes genomic instability.
- **H3K27me3 (written by EZH2)** represses genes; EZH2 is raised in mCRPC and predicts PSA recurrence.
- MicroRNA (18–25 nt) silences **many** mRNAs (**miR-371a-3p** in testis cancer) and siRNA **one**; long ncRNA HOTAIR drives castration resistance.

8 · HYPOXIA, VHL AND ANGIOGENESIS

ALGORITHM

1. In normoxia, HIF-α is prolyl-hydroxylated.
2. VHL (an E3 ubiquitin ligase) binds it for proteasomal degradation.
3. With hypoxia or VHL loss, HIF-α accumulates and dimerises with HIF-1β.
4. This switches on VEGF, angiopoietin and glycolytic genes.

- The **Warburg effect** (aerobic glycolysis) is best seen in clear cell RCC.

KEY POINT

VHL loss means permanent pseudohypoxia, driving angiogenesis, Warburg metabolism and immunosuppression; hence clear cell RCC responds to VEGF, HIF-2α and immune blockade.

- **Lineage plasticity** under driver targeting gives neuroendocrine prostate cancer.

9 · GU TUMOUR GENOMICS

Prostate

Gene	Significance
TMPRSS2–ERG	Commonest fusion (~half)
SPOP	Commonest mutated suppressor in primary tumours
TP53	Commonest alteration overall (more in advanced disease)
RB1, PTEN, TP53	Loss of ≥2 means aggressive variant, overlapping neuroendocrine
BRCA1/2	Germline plus somatic hit: give **PARP inhibitor**
ATM	Little PARP benefit
CDK12	Tandem duplications; poor response to hormones, chemo, PARP
MMR deficiency	Microsatellite instability (MSI): give immunotherapy
AR amplification	Late, under AR-targeted pressure; causes ADT resistance

Kidney

Subtype	Genomic hallmark	Therapy link
Clear cell (75–80%)	**3p loss** plus VHL, PBRM1, BAP1, SETD2	VEGF tyrosine kinase inhibitors (TKIs); belzutifan (HIF-2α)
Papillary (15–20%)	MET (7q); FH loss in subset	MET-active TKIs
Chromophobe	Mitochondrial DNA, mTOR pathway	mTOR-directed rational
Translocation	MiT/TFE fusions; young women	No targeted therapy
Sarcomatoid (any)	BAP1, CDKN2A loss	PD-L1-positive, so immunotherapy-responsive

- In clear cell RCC, VHL is mutated in 60–70%. **PBRM1** (~40%) means a better outcome and **BAP1** or **SETD2** (~10% each) a worse one.

Bladder (urothelial carcinoma, UC)

- UC has a high (APOBEC) mutation load, with TP53 most frequent.
- Of six consensus subtypes, **basal/squamous is largest (35%)**, luminal papillary 24%, neuroendocrine-like smallest 3%.
- **FGFR3** is altered in 15–20% of advanced UC and up to 37% of upper tract UC. These T-cell-poor tumours respond to **erdafitinib**, so test all advanced UC. HER2 is altered in 9–12% of MIBC.

Testis

- Testis cancer shows **12p gain** (†isochromosome 12p) and KIT/KRAS in seminoma; miR-371a-3p has >90% sensitivity and specificity.

Hereditary syndromes

P2 P1 *Hereditary urological malignancies / Familial malignancies in urology — VHL, HPRC (MET), HLRCC (FH), BHD (FLCN), BRCA2 > BRCA1, Lynch, HOXB13*

Not in the paper answer:

- VHL carries a >70% lifetime risk of clear cell RCC (belzutifan). HLRCC gives papillary type 2 RCC, Lynch gives UC and prostate cancer, and HOXB13 carriers need earlier PSA screening.

> **KEY POINT**
> The germline syndrome names the somatic driver (VHL: clear cell; MET: papillary; FH: type 2; BRCA: prostate), and the driver names the drug (HIF-2α, MET TKI, PARP, FGFR inhibitor, immunotherapy for MMR loss).

10 · CLINICAL NGS AND BIOMARKERS

- **Tumour-only** panels cannot separate somatic from germline variants.
- **Variant allele frequency (VAF)** depends on zygosity, copy number and purity; a heterozygous variant at 60% purity gives 30%. VAF of ~50% or ~100% suggests a **germline** variant, and single-digit VAF in cell-free DNA suggests **clonal haematopoiesis**.
- Circulating tumour DNA after treatment means minimal residual disease.
- A predictive marker needs a **biomarker–treatment interaction test** in a randomised trial; crossing lines mean predictive, parallel lines prognostic.
- Absolute benefit scales with baseline risk: a 50% relative reduction gives 15% absolute benefit at 30% risk but 1.5% at 3%.
- **HSD3B1 (1245A>C)** raises DHT synthesis, causing ADT resistance but abiraterone sensitivity.

11 · LANDMARK TRIALS

Trial	Design	Result	Take-home
†PROfound	Olaparib vs enzalutamide/abiraterone, HR-altered mCRPC	Better radiographic PFS and overall survival, driven by BRCA1/2	Germline and somatic HR testing in metastatic prostate cancer
†THOR	Erdafitinib vs chemotherapy, FGFR-altered metastatic UC after immunotherapy	Longer overall survival	FGFR testing of all advanced UC

55 Management of Abnormalities of the External Genitalia in Males

1 · NORMAL DEVELOPMENT AND PENILE SIZE

- Male differentiation (weeks 9–13) needs Leydig testosterone, 5α-reductase conversion to DHT and a functional androgen receptor; any break causes undervirilisation.
- The genital tubercle forms the glans, urethral folds the shaft, and labioscrotal swellings the scrotum.
- **Stretched penile length (SPL)** runs from pubis (fat pad compressed) to glans tip. At term it is **3.5 ± 0.7 cm**; in adults 13.3 ± 1.6 cm.
- The micropenis cut-off is −2.5 SD: **1.9 cm** from term to 5 months, 1.5 cm at 30 weeks, 2.0 cm at 34 weeks, ≈ 3.7–3.9 cm at 5–11 years and 9.3 cm in adults. Compare with the cut-off, not the mean.
- Minipuberty (first 3 months) gives a second phase of penile growth. In Tanner staging, testicular change comes first (stage 2); semen analysis is done at Tanner 5.

2 · PREPUCE: PHIMOSIS AND PARAPHIMOSIS

- **Physiological phimosis** is normal at birth: ≈ **90%** are **retractable by 3 years**, and < 1% remain phimotic at 17 years. Forceful retraction causes cicatricial phimosis.
- Treat persistent phimosis, balanitis, BXO or UTI.
- **Topical steroid**: betamethasone 0.05% twice daily for 30 days (or thrice daily for 21 days). About 1 in 6 fail.
- A retractile prepuce halves recurrent febrile UTI risk. The commonest acquired "penile lump" is **entrapped smegma**.
- **Paraphimosis** causes oedema within hours and may progress to gangrene.

ALGORITHM

1. Compress the glans with distal traction; this succeeds in most.
2. If it fails, reduce oedema: iced glove 5 min, granulated sugar 1–2 h, or punctures.
3. If still irreducible, do a dorsal slit, then circumcision.

3 · CIRCUMCISION

- The AAP (2012) found the benefits insufficient to recommend routine neonatal circumcision, but enough to justify access and payment.
- Before using a device (Gomco, Mogen, Plastibell), inspect the meatus to exclude hypospadias and megameatus. Older children need freehand surgery under GA.
- Analgesia is mandatory. **Dorsal penile nerve block is superior to EMLA**, whose prilocaine can cause methaemoglobinaemia.

Defer / avoid neonatal circumcision	Reason
Hypospadias, curvature, torsion, dorsal hood, buried or webbed penis	Prepuce needed for reconstruction
Large hydrocele / inguinal hernia	Skin pushed forward, causing secondary phimosis or a trapped penis
Coagulopathy (contraindication); hyperbilirubinaemia (some)	Haemorrhage
NICU setting	More complications

- **Benefits**: penile cancer occurs almost only in uncircumcised men. Febrile infant UTI falls, but **111** circumcisions prevent one UTI.
- There is ≈ **60%** protection against HIV acquisition in men, but no reduction in transmission to partners. HSV-2 and high-risk HPV are also lower, and no sexual dysfunction is proven.

Complications

P1 *Circumcision undone — 0.2–5% (10–20× after infancy); bleeding commonest; adhesions/skin bridges*
(steroids don't help); trapped penis; revision under GA

Not in the paper answer:

- An amputated glans should be re-attached within **8 h**.
- Degloving heals by re-epithelialisation, with **no immediate suturing or graft**.
- A skin bridge can cause chordee or torsion, so excise it.

> **KEY POINT**
> Late sequelae are problems of skin quantity (too much removed: trapped penis, chordee, torsion; too little: redundancy, adhesions) or of the newly exposed meatus (stenosis).

Meatal stenosis and BXO

P2 P1 *Management of Urethral meatal stenosis / Lichen sclerosus (BXO) — normal calibre by age, post-circumcision meatitis, upward stream, ventral meatotomy; BXO features, steroid, circumcision, buccal*
mucosa

Not in the paper answer:

- Meatal stenosis occurs almost exclusively in **circumcised** boys.
- BXO in boys is rare under 5 years. Steroids help only mild preputial disease, so **circumcision is preferred**, with or without meatoplasty.
- About 1 in 5 with meatal BXO need further meatal surgery.

4 · INCONSPICUOUS PENIS VERSUS MICROPENIS

> **KEY POINT**
> **Stretched length is the single discriminator.** An inconspicuous penis (buried, webbed, trapped) has normal SPL hidden by skin, fat or scar, so treatment is anatomical. A micropenis is small but normally formed, so it needs endocrine work-up.

Buried penis type	Mechanism	Treatment
Congenital	Inelastic dartos; skin not anchored to deep fascia	Surgery
Obesity	Suprapubic fat hides shaft	Weight loss; liposuction only in adolescents
Trapped (post-circumcision)	Cicatrix embeds penis in fat	Dilation or betamethasone with retraction; then repair under GA at ≥ 6 months

- If retracting skin lateral to the shaft exposes the glans easily, the shaft is normal.
- **Repair** excises the dysgenic dartos bands, fixes dorsal subcutaneous tissue to pubic fascia and scrotal tissue to the ventral base, and covers with unfurled prepuce, Z-plasty, flap or graft.
- **Webbed penis** (penoscrotal fusion) is repaired by transverse incision with vertical closure, or Byars' flap. Avoid neonatal circumcision.

Micropenis

- Micropenis is normally formed but ≥ 2.5 SD below the mean (< **1.9 cm** at term). Testes are often small or undescended.
- A defect before week 12 (placental hCG) gives a small penis with hypospadias; gonadotrophin deficiency after 14 weeks gives a normally formed micropenis.

Cause	Examples	Hormones
Hypogonadotrophic hypogonadism (commonest)	Kallmann, Prader–Willi, Laurence–Moon–Biedl, CHARGE, hypopituitarism, midline brain defects	Low LH and FSH

Cause	Examples	Hormones
Hypergonadotrophic	Rudimentary testes, gonadal dysgenesis, Robinow	High LH/FSH; no testosterone rise after hCG
Idiopathic	Delayed gonadotrophin timing	Normal axis
Partial androgen insensitivity (rare)	—	Ambiguity commoner

ALGORITHM

1. History, examination, SPL and **karyotype**.
2. Refer to a paediatric endocrinologist; measure testosterone before and after hCG, with or without a GnRH test.
3. Screen the anterior pituitary (glucose, Na/K, cortisol, thyroid); then MRI brain.
4. Test **end-organ response: intramuscular testosterone enanthate × 3 months**. Short courses spare height; prolonged ones advance bone age. Repeat at puberty.

- Most achieve male identity, standing voiding and satisfactory sexual function.

5 · CURVATURE (CHORDEE WITHOUT HYPOSPADIAS) AND TORSION

P1 *Congenital Penile curvature — Devine–Horton types, artificial erection, plication/septotomy/graft*

Not in the paper answer:

- Repair if sexual dysfunction is expected, with concurrent penile surgery, or if growth will worsen it.

Type	Associations	Repair (stepwise)
Ventral (commonest)	Hypospadias; dorsal hood; a prominent frenulum causes distal chordee	Deglove; excise fibrous tissue superficial to Buck's; Byars' flaps; if still curved, dorsal plication, Nesbit or corporal rotation; if the urethra is short, reconstruct it; if frenular, frenulotomy
Dorsal	Isolated, epispadias	Ventral plication; protect dorsal neurovascular bundles
Lateral	Corporal asymmetry; straight when flaccid, noticed on erection	Plicate corpus **opposite** maximal curvature

- Work superficial to deep. An **artificial erection** locates the apex and confirms correction.
- **Penile torsion** is usually **counter-clockwise (to the left)**, < 60° and skin-level. If < 90°, deglove and rotate; if ≥ 90°, also divide basal bands and place non-absorbable anchoring sutures at the corporal base.

6 · ABNORMAL NUMBER, MASSES AND LYMPHOEDEMA

Feature	Aphallia	Diphallia
Incidence / defect	1 in 10–30 million; genital tubercle fails; 46,XY, normal scrotum and testes	1 in 5 million; from accessory to complete duplication
Associations	Cryptorchidism, VUR, renal agenesis, horseshoe kidney, imperforate anus, cardiopulmonary; ≈ 1/3 stillbirth/neonatal death	Hypospadias, bifid scrotum, bladder duplication, renal, pubic diastasis, anal, cardiac
Work-up / decision	Karyotype, anomaly screen, MRI; then neophallus or reassignment (reassigned may develop male identity)	Renal US, VCUG, penile US/MRI; individualised

- In aphallia the urethra most often opens **postsphincteric** (anal verge), with the best survival.
- Penile cysts are commonest and are excised; **juvenile xanthogranuloma is observed for 1 year**.
- Lymphoedema may be primary (Milroy, Meige, praecox) or acquired (anogenital Crohn disease); surgery is excisional, not drainage. Haemangiomas involute, while vascular malformations persist

and are resected.

7 · URETHRAL ANOMALIES

P2 *Megameatus intact prepuce / Anterior urethral valve — MIP: wide meatus under complete prepuce, found at circumcision, pyramid repair; AUV: bulbar, diverticulum lip, VCUG, endoscopic ablation*

Not in the paper answer:

- If MIP is recognised during circumcision, stop and preserve the prepuce.
- **Urethral duplication** is usually sagittal (≈ 94%), and the **ventral channel is the functional one**. The **Effmann** classification is type I incomplete (IA distal, IB proximal, blind), type II complete (IIA two meatuses, including IIA2 "Y-type" perineal; IIB one meatus), and type III with bladder duplication.
- Duplication presents with double stream, incontinence or UTI and is evaluated by VCUG, RUG and cystoscopy. **Never use the hypoplastic dorsal channel as the main urethra**.
- **Congenital urethral fistula** is subcoronal with a normal meatus and is closed in layers.
- **Cowper syringocele** (Maizels types: simple, perforated, imperforate, ruptured) needs endoscopic unroofing if symptomatic.
- **Megalourethra** is **scaphoid** (commoner; spongiosum deficient) or **fusiform** (rarer; cavernosa also deficient, often lethal). Both go with prune-belly and are repaired by reduction urethroplasty.

8 · PRIAPISM

P3 *Priapism: Diagnosis and Management — ischaemic vs non-ischaemic vs stuttering, blood gas, aspiration/phenylephrine, distal then proximal shunts, embolisation*

Type (children)	Cause	Treatment
Ischaemic	**Sickle cell** (commonest), leukaemia	Hydration, O_2, alkalinisation, analgesia, with or without transfusion; then aspiration with or without phenylephrine; then shunt
Non-ischaemic	Straddle injury causing an arteriocavernous fistula	Observe; if persistent, superselective embolisation
Stuttering	Sickle cell	Bedtime pseudoephedrine; oral β-agonist; long-term PDE5 inhibitor
Neonatal	Idiopathic, birth trauma, polycythaemia	None — resolves in 2–6 days

- Corporal blood gas decides urgency: ischaemic blood is dark, hypoxic and acidotic in a rigid, painful penis, while arterial blood is bright red in a non-rigid, painless one. Counsel about fibrosis and ED.

9 · SCROTAL ANOMALIES

- **Penoscrotal transposition** is failed inferomedial migration of the labioscrotal swellings. It goes with proximal hypospadias, and in the complete form **up to 75%** have upper-tract anomalies, so do renal US and VCUG.
- Repair it with the hypospadias by **rotation-flap scrotoplasty**.

Anomaly	Feature	Association	Management
Bifid scrotum	Unfused folds	Proximal hypospadias	Scrotoplasty with hypospadias repair

Anomaly	Feature	Association	Management
Ectopic scrotum	Suprainguinal (commonest), infrainguinal, perineal	Ipsilateral renal agenesis/dysplasia, ectopic ureter; cryptorchidism; perineal lipoma usually isolated	Upper-tract US; scrotoplasty and orchidopexy at 6–12 months
Hypoplasia / agenesis	Under-developed / absent	Undescended testis (UDT); agenesis — all XY, bilateral UDT	—
Scrotoschisis	Rupture, extruded testis	Meconium peritonitis	Explore, orchidopexy, close

10 · INGUINAL HERNIA AND HYDROCELE

*All reflect incomplete closure of the **processus vaginalis**.*

Entity	Processus vaginalis	Plan
Indirect hernia	Patent; contents enter	Repair soon after diagnosis
Communicating hydrocele	Patent, fluid only	Repair if > 1–2 y, enlarging or fluctuating
Hydrocele of cord	Isolated patent segment	Rarely resolves; repair after 1 year
Scrotal (non-communicating)	Closed	Observe — most resolve by 1 y
Abdominoscrotal	Closed; through ring (≈ 1%; bilateral 30%)	Excise/drain; ring tightening unnecessary

- Hernia affects 1–5% of children and more with prematurity (≈ 30% if < 1 kg); right to left is 3:1. Bilateral risk is higher with prematurity, age under 1 year, cryptorchidism, ascites and VP shunts.
- An **intermittent bulge** means a patent processus, so operate. Incarceration needs immediate exploration, and reducible hernias elective repair within weeks; in preterm infants repair before discharge.

ALGORITHM

1. Skin-crease incision; open the external oblique, sparing the ilioinguinal nerve (not opened in young infants, Mitchell-Banks).
2. Split the cremaster and find the sac anteromedially; separate it from the vas and vessels.
3. Perform **high ligation** at the internal ring and tighten a wide ring.
4. Finally, **confirm the testis is scrotal**.

- Recurrence is 0.5–1% (3–6% after incarceration), from a missed or torn sac or a missed direct hernia. Other complications are iatrogenic cryptorchidism, atrophy and vas injury.
- Laparoscopic repair now matches open repair and closes the contralateral ring.
- Contralateral patency is ≈ 40% in infants, but a metachronous hernia develops in only ≈ **7%**, so routine contralateral exploration is not advised.

11 · ACUTE SCROTUM

Only cord torsion is time-critical.

Feature	Cord torsion	Appendage torsion	Epididymitis
Frequency	20–30%	**40–60% (commonest)**	5–15%
Peak age	Perinatal; 12–16 y	7–12 y	Infancy, puberty
Onset / vomiting	Acute severe; vomiting common	Sudden or insidious; uncommon	Insidious; rare
Cremasteric reflex	Usually absent	Present	Present

Feature	Cord torsion	Appendage torsion	Epididymitis
Signs	High-riding, horizontal lie	**Blue dot**, upper-pole nodule	Tender epididymis; pyuria 20–40%
Colour Doppler US (CDUS)	Absent/reduced flow; **whirlpool/snail sign**	Avascular upper-pole lesion, hyperaemia	Increased epididymal flow
Management	**Emergency exploration**	Ice, NSAIDs, rest	Supportive; antibiotics only if UTI

Intravaginal torsion

- The **bell-clapper** deformity is found in ≈ 12% at autopsy. Torsion affects ≈ 1 in 4000, is familial and is commoner if cryptorchid. Delay to presentation is the main cause of testis loss.
- A cremasteric reflex is present in up to 10% of torsions, so it does not exclude torsion.
- **TWIST score**: †swelling 2, hard testis 2, absent reflex 1, nausea/vomiting 1, high-riding 1. A score ≥ 5 is high risk, so go to theatre without imaging; ≤ 2 is low risk.
- On CDUS, absent flow is specific but only 63–90% sensitive. **Do not delay surgery for imaging when suspicion is high.**

Hours from onset	0–6	7–12	13–18	19–24	> 24	> 48
Orchidectomy risk	**≈ 5%**	20%	40%	60%	80%	90%

ALGORITHM

1. If torsion is suspected, explore the scrotum.
2. Detorse and apply warm packs. **Meanwhile fix the contralateral testis with non-absorbable suture**.
3. Reassess. If viable, then fix in a dartos pouch or with non-absorbable sutures. If non-viable, then do an orchidectomy; offer a prosthesis later.

- Manual detorsion fails in ≈ 1/3, and orchidopexy is still needed after success.
- Salvaged testes may still atrophy, with semen defects from ischaemia–reperfusion.
- **Intermittent torsion** precedes acute torsion in 30–50%, so offer **elective bilateral orchidopexy**.

Perinatal (extravaginal) torsion

- Cord and tunica twist together before scrotal fixation, giving a hard painless testis. Salvage is rare and it is bilateral in 5–22%; explore urgently if a normal postnatal exam was documented.

Appendage torsion, epididymitis, others

- Explore appendage torsion only if cord torsion is not excluded or pain is prolonged.
- Treat epididymitis with ice, NSAIDs and elevation, adding antibiotics if pyuria. A prepubertal boy with a positive culture needs renal US and VCUG.
- Idiopathic scrotal oedema shows the **fountain sign** and is treated with an antihistamine.

12 · ADOLESCENT VARICOCELE

P1 P3 *Varicocele – Pathophysiology / Varicocele: controversies — left-sided anatomy, nutcracker, grading, right/non-decompressing: image for mass; adolescent: observe, repair for asymmetry*

Not in the paper answer:

- Prevalence is ≈ **15%**; it appears after 10 years and peaks at Tanner 3.
- Measure testicular volume by **US (L × W × H × 0.71)**, serially. Peak retrograde flow > 38 cm/s correlates with asymmetry.
- Catch-up growth may be spontaneous, so **observation is the default**.

KEY POINT

Indications for repair: left testis > **20%** smaller (or bilateral hypotrophy), pain, abnormal semen (Tanner 5 / ≥ 18 y), or intratesticular varicocele with asymmetry. Total volume < 30 mL and low total motile count predict poor semen.

Technique	Recurrence	Hydrocele	Note
Open retroperitoneal (Palomo, mass ligation)	2–4%	Up to 30%	Lymphatics ligated
Laparoscopic, non-sparing	0–9%	11–32%	—
Laparoscopic artery/lymphatic-sparing	1–7%	0–4%	Lymphatic sparing cuts hydrocele
Microscopic subinguinal	0–10%	0–6%	Best balance; rare atrophy
Non-microscopic inguinal	7–33%	8–14%	—
Sclerotherapy (retrograde/antegrade)	6–35% / 4–12%	Occasional	Mainly for recurrence

- Hydroceles may appear 2 or more years after repair, and up to half need intervention.

13 · EPIDIDYMAL AND VASAL ANOMALIES

- Spermatoceles are postpubertal, contain sperm and are mostly observed.
- Bilateral absent vas suggests a **CFTR** mutation; unilateral absence suggests **renal anomalies**.

58 Management of Abnormalities of the Genitalia in Girls

1 · EMBRYOLOGY THAT EXPLAINS THE ANOMALIES

- The **perineal body** splits the cloacal sphincter, so urogenital diaphragm and external anal sphincter are **both pudendal-innervated**.
- Müllerian ducts arise lateral to wolffian ducts, so a **müllerian anomaly travels with an ipsilateral renal anomaly**. Fused ducts form uterus, cervix and upper two-thirds of vagina; **sinovaginal bulbs** form the distal vagina.
- **Caudal regression** (weeks 4–5 insult) is linked to maternal diabetes; cranial extension gives VATER or MURCS (müllerian aplasia, renal aplasia, cervicothoracic somite dysplasia).

2 · EVALUATION AND CLASSIFICATION

- The **pull-down manoeuvre** (labia drawn caudally and laterally) shows the distal vagina. Discharge or bleeding needs vaginoscopy under anaesthesia.
- Ultrasonography (US) shows anatomy and upper tracts; **magnetic resonance imaging (MRI) is the gold standard** for müllerian anatomy.

Category	Mechanism	Examples
Hypoplasia/agenesis	Fails to form	Mayer–Rokitansky–Küster–Hauser (MRKH)
Vertical fusion	Faulty müllerian–sinus canalisation	Imperforate hymen, transverse septum, distal atresia
Lateral fusion	Failed midline fusion/resorption	Didelphys, Herlyn–Werner–Wunderlich syndrome (HWWS)

KEY POINT

Decision hinge: **is it obstructive?** Obstruction needs prompt drainage; non-obstructive anomalies wait until intercourse or fertility demands surgery.

3 · CLITORIS

- **Clitoromegaly** is commonest from **congenital adrenal hyperplasia (CAH)**, so check electrolytes, 17-hydroxyprogesterone and karyotype (Ch 59). A large hood with normal corpora is **pseudo-enlargement**; a **diminutive clitoris** suggests complete androgen insensitivity syndrome (CAIS).

4 · VESTIBULE: THE INTERLABIAL MASS

- **Urethral prolapse** affects prepubertal African-ancestry girls: a **doughnut with central meatus** needing no imaging. Sitz baths and topical steroid or oestrogen come first, then **circumferential excision**.
- A **Skene cyst** gives a **displaced meatus, eccentric stream** and usually ruptures.
- A **Gartner cyst / covered ectopic ureter** on the anteromedial vaginal wall is incised; upper-pole heminephrectomy follows if wetting or infection persists.
- A **prolapsed ureterocele** gives an infant retention and a pink-to-dusky mass; decompress, then reduce.
- An **ectopic ureter** below the sphincter causes **daytime wetting despite normal voiding**.

> **KEY POINT**
>
> Read every mass against the meatus: central (prolapse), displaced (Skene), separate/vaginal (Gartner, ectopic ureter, hymen, tumour). Any cyst of urethral or vaginal origin needs a **renal US, always**.

- A **labial swelling** is commonest an inguinal hernia; **childhood asymmetric labium majus enlargement (CALME)** is physiological and is observed.

5 · VAGINA: OBSTRUCTION AND ABSENCE

Block level and uterus presence decide the operation.

Entity	Level / uterus	Clue	Treatment
Imperforate hymen	Hymen; normal	Whitish bulge (neonate); bluish bulge and amenorrhoea (adolescent)	**Transverse** incision; not aspiration
Transverse septum	Upper/middle third; normal	Normal hymen, blind vagina	Excision, Z-plasty, balloon; hollow mould
Distal atresia	Sinus; normal	Shallow dimple; distended vagina per rectum	Pull-through with or without flaps
MRKH	Bulbs/plate; absent	Hymenal fringe and pouch; 46,XX	Dilation first
Cervical agenesis	Cervix; uterus present	Obstruction; MRI	Often hysterectomy and neovagina
HWWS	Lateral fusion; didelphys	Menstruates; lateral bluish bulge	Wide septal incision

- **Imperforate hymen** is the commonest obstructive anomaly; aspiration causes ascending infection.
- **Transverse septum** (~1 in 70,000): delay causes endometriosis, and **MRI distinguishes a high septum from cervical agenesis**. **Z-plasty** makes the scar longitudinal; the main complication is **stenosis**, and a **hollow mould** is used when a uterus is present.
- **Distal atresia**: the haematocolpos acts as a tissue expander; drain, then pull through.

MRKH (müllerian agenesis)

- MRKH (~1 in 5000) is 46,XX with **functioning ovaries**, the second commonest primary amenorrhoea after gonadal dysgenesis.

- Renal anomaly occurs in ≈ **one-third**, skeletal in 10–20%. **Type A** is symmetrical without extra-müllerian anomaly; **Type B** is asymmetrical with renal, Klippel–Feil or MURCS anomalies and needs full renal and skeletal work-up.

6 · VAGINAL REPLACEMENT

- **Dilation is first-line** (Frank; Ingram); surgery is no better. **Vecchietti** uses an olive with laparoscopic traction.
- **McIndoe** places a split-thickness graft over a stent; it stenoses, so dilation is mandatory. Alternatives include **buccal mucosa** and **robotic peritoneal vaginoplasty**.
- **Sigmoid** is the default bowel; excise ~3 cm of intervening bowel to avoid overlapping suture lines and fix it against prolapse. **Ileum** stenoses more and is kept for an irradiated pelvis or absent colon.
- Bowel risks discharge and **adenocarcinoma**; skin risks stenosis and **squamous carcinoma**. Every neovagina needs **annual examination**.

> **KEY POINT**
> **Cervical atresia**: a perineum-to-uterus anastomosis loses the endocervical barrier and risks life-threatening ascending infection. Hysterectomy with neovagina is often best; otherwise use endometrial suppression and IVF with transmyometrial transfer.

7 · LATERAL FUSION ANOMALIES

HWWS girls menstruate, so obstruction is missed.

- **Didelphys** has a septate vagina in up to 75%; excise it only for symptoms. Incise a **septate uterus** only for recurrent pregnancy loss.
- **HWWS** is didelphys with a unilateral imperforate hemivagina: pain **without amenorrhoea**, a lateral bluish bulge and **ipsilateral renal agenesis**. Treat by wide septal incision.

8 · ACQUIRED DISORDERS

- **Vaginal rhabdomyosarcoma** is a grapelike bleeding mass under 2 years with the best prognosis of genital primaries. Chemotherapy is primary; IRS trials cut surgery from **100% to 13%**.
- **Labial adhesions** are the commonest interlabial abnormality (peak 13–23 months); **80% resolve in 1 year**, so treat only symptoms. Use topical oestrogen or betamethasone 0.05% before separation.

P1 *Female circumcision — FGM ~200 million, WHO types I–IV (III = infibulation), complications, midline deinfibulation*

- **Condylomata** suggest abuse, though perinatal transmission occurs; over 50% resolve.
- An **inguinal hernia** may present CAIS (~1–2.5%); the **round ligament is absent in genetic males**. If CAIS is found, preserve the testes.

59 Differences of Sexual Development

1 · DEFINITION AND NOMENCLATURE

- In a **difference of sex development (DSD)** chromosomal, gonadal or anatomical sex is atypical. The **2006 Chicago consensus** replaced "intersex" and "hermaphrodite" with karyotype-based terms.

Old term	2006 term
Male pseudohermaphrodite	46,XY DSD

Old term	2006 term
Female pseudohermaphrodite	46,XX DSD
True hermaphrodite	Ovotesticular DSD
XX male / XX sex reversal	46,XX testicular DSD
XY sex reversal	46,XY complete gonadal dysgenesis (not androgen insensitivity syndrome, AIS) MCQ

- Offer endocrine and genetic testing for **proximal hypospadias even with descended testes**.

2 · NORMAL SEX DEVELOPMENT

- **SRY** upregulates **SOX9**, making Sertoli cells. Genes acting elsewhere explain syndromic DSD: **WT1** (Wilms tumour, nephropathy), **SF1** and **DAX1** (adrenal failure), SOX9 (campomelic dysplasia).
- **Sertoli anti-Müllerian hormone (AMH)** from 7–8 weeks makes Müllerian ducts regress by ≈ 10 weeks. **Leydig testosterone** from ≈ 9 weeks builds the Wolffian ducts.
- **5α-reductase 2 makes dihydrotestosterone (DHT)**, the active androgen for prostate, urogenital sinus and external genitalia.

> **KEY POINT**
> The testis uses two separate hormones: AMH removes Müllerian ducts, testosterone/DHT builds Wolffian and external structures. A uterus in 46,XY means Sertoli (AMH) failure, that is dysgenesis, not a synthesis or receptor defect.

3 · CLASSIFICATION (2006 / CHICAGO)

Category	Conditions
Sex chromosome DSD	47,XXY Klinefelter; 45,X Turner; 45,X/46,XY mixed gonadal dysgenesis (MGD); 46,XX/46,XY chimerism
46,XX DSD — gonadal	Gonadal dysgenesis; ovotesticular DSD; testicular DSD (SRY+, SOX9 duplication)
46,XX — androgen excess	Fetal: congenital adrenal hyperplasia (CAH: 21-hydroxylase, 11β-hydroxylase, 3β-HSD, P450 oxidoreductase); fetoplacental: aromatase deficiency; maternal: luteoma, drugs
46,XX — other	Cloacal anomalies; Mayer–Rokitansky–Küster–Hauser (MRKH); vaginal atresia
46,XY DSD — gonadal	Complete/partial gonadal dysgenesis (SRY, SOX9, SF1, WT1, DHH, MAP3K1); ovotesticular; testis regression
46,XY — androgen synthesis	LH receptor; Smith–Lemli–Opitz; StAR; 3β-HSD; 17α-hydroxylase; 17β-HSD3; 5α-reductase 2
46,XY — androgen action	Complete and partial AIS (CAIS, PAIS)
46,XY — other	Persistent Müllerian duct syndrome (PMDS); vanishing testis; isolated hypospadias; hypogonadotropic hypogonadism; cryptorchidism

- 3β-HSD and P450 oxidoreductase deficiency virilise 46,XX yet undervirilise 46,XY.

4 · 46,XX DSD

CAH is commonest and can kill a newborn through salt loss.

CAH — 21-hydroxylase deficiency

- CAH is the commonest 46,XX DSD and **commonest cause of neonatal atypical genitalia**. 21-hydroxylase deficiency (21-OHD) is **95%** of CAH; it is autosomal recessive, ≈ 1 in 15,000.

- 46,XX infants have normal ovaries and Müllerian structures, often with a **common urogenital sinus**; some look like a boy with **bilateral impalpable gonads**. Virilisation is graded **Prader I–V**.
- **17-hydroxyprogesterone (17-OHP) is raised**; measure it after **36 h**.
- A **salt-wasting crisis** (low sodium, high potassium, acidosis, shock) usually comes no earlier than **day 5–7**.
- Treat with glucocorticoid, mineralocorticoid and salt (hydrocortisone ≈ 10–15 mg/m²/day, fludrocortisone, NaCl in infancy; double-to-triple the dose in illness or surgery).
- Affected boys look normal, so **newborn screening** matters most.
- Males may get **testicular adrenal rest tumours**, which are benign and regress with more glucocorticoid but obstruct tubules, so bank sperm.
- Forms are **classic salt-wasting**, **simple virilising** and **nonclassic** (20–50% enzyme activity; hirsutism, oligomenorrhoea; may need no treatment).

Enzyme	Share	BP / K^+	46,XY	Signature
21-hydroxylase	95%	Salt wasting; low BP, high K^+	Normal	High 17-OHP
11β-hydroxylase	≈ 5%	Hypertension, low K^+	Normal	—
3β-HSD	<1%	Salt wasting	Undervirilised	High dehydroepiandrosterone (DHEA)
17α-hydroxylase	Rare	Hypertension, low K^+	Female/atypical; delayed puberty	Low androgens

Other 46,XX DSD

P1 P2 *46-XX Male — SRY translocated to X; testes, no Müllerian structures; azoospermia (no AZF)*

- Aromatase deficiency virilises mother and infant; maternal androgen exposure ends at birth, so no postnatal therapy is needed.

5 · 46,XY DSD

A uterus separates dysgenesis from synthesis or receptor defects.

Condition	Key features
Complete gonadal dysgenesis (Swyer)	Streaks, female genitalia, uterus; absent puberty; tumour 30–45%, so prophylactic gonadectomy; oestrogen; donor-ovum pregnancy
Partial gonadal dysgenesis	Dysgenetic testes, atypical genitalia, variable Müllerian structures. WT1: Denys–Drash (congenital nephrotic syndrome, Wilms 90%) or Frasier (late focal segmental glomerulosclerosis, gonadoblastoma 60%, no Wilms)
CAIS	X-linked; female genitalia, short blind vagina, no uterus; breasts by aromatisation; ≈ 50% have inguinal hernia; 1–2% of girls with hernia are CAIS
PAIS	Atypical genitalia; identity mostly male; pubertal breast growth
5α-reductase 2	Autosomal recessive; atypical/female at birth, blind pouch; **virilises at puberty**, fertility reported; testosterone:DHT **>20:1**; "guevedoces"
17β-HSD3	High androstenedione:testosterone ratio; virilises at puberty
LH receptor defect	Leydig hypoplasia; no hCG response
PMDS	AMH or AMH-receptor mutation (autosomal recessive); male genitalia with uterus and tubes; found at hernia repair or orchidopexy; 30–50% of transverse testicular ectopia
Vanishing testes	Loss after first trimester: male genitalia, impalpable testes, no Müllerian structures; undetectable testosterone/AMH, high gonadotropins at mini-puberty; testosterone from adolescence

- PAIS raised male may respond to high-dose testosterone; raised female, use a GnRH agonist or gonadectomy, then oestrogen. In CAIS, vaginal dilation usually suffices.

- **PMDS (Clarnette)**: type 1 is bilateral intra-abdominal (60–70%); type 2, testis in sac or scrotum with contralateral hernia (20–30%); type 3, transverse ectopia (10%).
- PMDS needs orchidopexy. The vasa hug the uterus, so Müllerian malignancy risk (3–8%) is weighed against careful removal sparing the vasa.

6 · SEX CHROMOSOME DSD AND OVOTESTICULAR DSD

P4 P3 *Klinefelter's syndrome / Mixed gonadal dysgenesis — 47,XXY features, hormones, micro-TESE; MGD 45,X/46,XY asymmetry, second commonest atypical genitalia, tumour 15–35%, remove streaks,*

Turner screening

Not in the paper answer:

- **Turner with Y** material: tumour 12–20%; the 2017 guideline advises gonadectomy at diagnosis, on low-level evidence.
- **Turner**: coarctation, bicuspid valve, horseshoe kidney (renal anomaly ⅓–½) and ovarian insufficiency ≈ 95%, so give lifelong cardiac and renal screening, growth hormone and oestrogen.
- **Ovotesticular DSD** has ovarian and testicular tissue; most are 46,XX, with a sinus and uterus. A **vas always accompanies a testis**, a tube an ovary.
- The ovary is usually normal and the testis dysgenetic; tumour risk is ≈ 3%. Biopsy both poles of an ovotestis and defer irreversible steps.

7 · EVALUATION OF THE NEWBORN WITH ATYPICAL GENITALIA

P3 *DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — history, exam, gonads*

as key sign, karyotype/FISH, 17-OHP, 1-week hormone table, karyotype-based algorithm, sex of rearing, surgery timing, gonadal risk

Not in the paper answer:

- Late presenters include CAIS, Swyer, MRKH, Turner with Y, Klinefelter and 5α-reductase deficiency; after puberty, high LH and testosterone with female genitalia means CAIS.

> **KEY POINT**
>
> **Emergency**: atypical genitalia with bilateral impalpable gonads is salt-wasting CAH until proven otherwise. Check electrolytes and 17-OHP, and watch for crisis from the end of week 1. Designation and surgery are not emergencies.

- A multidisciplinary team (MDT) reviews within **24–48 h**. A urologist without a team transfers early.
- Count palpable gonads (ovaries do not descend): none suggests CAH; one, MGD, ovotesticular DSD, PAIS or partial dysgenesis; two, 46,XY DSD.
- Hypospadias with undescended testis (Kaefer 1999): DSD ≈ 30%; 50% if testis impalpable; 65% if meatus proximal.

ALGORITHM

1. Send karyotype (≥45 cells, for mosaicism; FISH for X/Y in hours), electrolytes, and 17-OHP at ≈ 36 h.
2. Pelvic ultrasonography for uterus and gonads.
3. At ≈ 1 week: LH, FSH, testosterone, oestrogen, androstenedione, DHT, AMH, with or without hCG stimulation.
4. Genetic panel, microarray or exome.
5. If still unclear, then MRI, genitography, endoscopy, laparoscopy with or without biopsy (**no gonadectomy during diagnosis**).

- Normal AMH means functioning Sertoli (or granulosa) cells; high LH and FSH mean gonadal failure.
- A precise diagnosis is nearly always made in 46,XX DSD but in **<50%** of 46,XY.

8 · GENDER ASSIGNMENT AND MDT

- The **Money (1975)** model of early designation, early surgery and secrecy failed; shared decisions replaced it.
- Female trend: 46,XX CAH, CAIS, Swyer, LH receptor defect. Male trend: PAIS, 5α-reductase, 17β-HSD3, partial dysgenesis.
- Dysphoria is highest in 5α-reductase deficiency raised female (≈ half) and lowest in 46,XX CAH raised female.
- Micropenis (<2.5 SD; <2 cm at 0–5 months) needs an endocrine work-up; female reassignment is no longer recommended.

9 · SURGERY

- **Feminising** surgery (Ch 60): clitoral surgery is irreversible and infant benefit is unproven. Ideally defer it until the individual can take part.
- **Masculinising** surgery (proximal hypospadias, orchidopexy) has high complications but is revisable, so infant repair is often chosen.
- In MRKH, dilation comes first. Uterine remnants are removed or suppressed, **never joined to a neovagina** (no cervix).

10 · GONADAL TUMOUR RISK

Risk rises with Y material (gonadoblastoma locus, TSPY), dysgenesis and intra-abdominal position, and falls with a scrotal gonad.

Risk	Condition	Germ cell tumour	Management
High	Dysgenesis with Y, intra-abdominal (Swyer, MGD)	15–35% (Swyer 30–45%)	Gonadectomy at diagnosis
	Frasier	60%	Gonadectomy
	Denys–Drash	40%	Gonadectomy
	PAIS, non-scrotal	50%	Gonadectomy
Intermediate	Turner + Y	12%	Gonadectomy (debated)
	17β-HSD3	28%	Monitor or gonadectomy
	PAIS / dysgenesis, scrotal	Unknown	Biopsy at puberty (≥30 tubules, OCT3/4)
Low	CAIS	2%	Keep until after breast development
	Ovotesticular DSD	3%	Remove discordant testicular tissue
?None	5α-reductase; Leydig hypoplasia	Unknown	Unresolved

11 · ETHICS

- Respect autonomy, be transparent, and avoid irreversible procedures before consent unless necessary. Psychology is integral.

60 Surgical Management of Differences of Sexual Differentiation and Cloacal and Anorectal Malformations

1 · PRINCIPLES

P3 *DSD: Evaluation and Management of the Newborn with Ambiguous Genitalia — classification, work-up, CAH care, team, sex of rearing, gonadal risk*

Not in the paper answer:

- Endocrine Society 2018: families with experienced surgeons decide. Never use a flap for a high confluence. Introitoplasty is needed after puberty in 25–100%.

2 · UROGENITAL SINUS AND CLOACA — CLASSIFICATION

In a **urogenital sinus** the vagina still communicates with the urinary tract (usually the mid-distal urethra), and both exit as one **common channel**.

Entity	Distinguishing feature
Genital atypia states	Commonest; mostly congenital adrenal hyperplasia (CAH)
Pure urogenital sinus	Normal external genitalia; behaves like a cloaca
Persistent cloaca	Rectum also joins, so there is **one perineal opening**
Female exstrophy	Probably a form of urogenital sinus

- CAH causes half of all newborn atypical genitalia and **90%** of 46,XX cases. Salt wasting occurs in **67–75%**; get a karyotype within **48 h**.

> **KEY POINT**
>
> **Governing principle**: **true urethral length** (bladder neck to confluence) decides the operation far more than common-channel length, in sinus and cloaca. Confluence level is a continuum.

- **Ganesan concept** MCQ: in CAH the proximal urethra is normal (bladder neck to confluence **1.5–2 cm** in infants). Virilisation lengthens the **vagina-to-meatus** distance, so a "high" CAH confluence is an illusion.

3 · CLINICAL EVALUATION

- The **first threat is metabolic**: monitor electrolytes for salt wasting before reconstruction. **Hypertension** suggests 11β-hydroxylase deficiency.
- A **suprapubic mass** (bladder, **hydrometrocolpos**, or both) is often the only early sign of a pure sinus.
- The **Huffman index** (glans width × length) is normally **<3.5 mm²**; **>10 mm²** is a concern.
- **Hydrocolpos in cloaca** forms because urine refluxes into poorly draining vagina(s). It is commoner with a common channel **>3 cm**, duplication and posterior cloaca.

Associated (cloaca)	Findings	Frequency
Renal	Solitary kidney, dysplasia, pelviureteric junction obstruction, ectopia	Majority
Uterovaginal	Duplication/septation	**53–67%**
Spinal	Sacral anomalies; tethered cord ≈ one-third (mostly no skin stigmata)	Cord anomaly on magnetic resonance imaging (MRI) ≤ 43%
VACTERL; gut	Tracheo-oesophageal fistula; malrotation, duodenal atresia	Correct malrotation at colostomy

> **KEY POINT**
>
> Two early dangers: **adrenal crisis** (CAH) and **obstructive hydrocolpos** (pure sinus, cloaca). The number of perineal orifices tells you which world you are in.

4 · IMAGING AND ENDOSCOPY

CAH needs little imaging; a cloaca needs a full 3-D map.

Modality	CAH sinus	Pure sinus / cloaca
Ultrasonography (US)	Mandatory; **cerebriform adrenals**	Mandatory for kidneys; poor for Müllerian anatomy
Genitography	Least useful; often inaccurate	**Mandatory**, as a computed tomography (CT) or MRI cloacogram
MRI	Limited	Pelvis, spine, sphincter muscles
Endoscopy	At reconstruction	Early, separate: decompress and define

- On a **genitogram**, a **cervical impression** at the vaginal dome means female internal organs.
- Do a **gonadal biopsy** only if it would change the gender of rearing, using a **deep longitudinal incision** to reach testicular tissue within an ovotestis.
- **Prenatal cloaca** shows fetal ascites, bilobed pelvic cysts and hydronephrosis. **Always do echocardiography.**
- In duplicated vaginas the rectal fistula most often opens **within the vaginal septum**. At **early endoscopy**, **empty the vagina first**.

Common channel (Peña)	**<3 cm**	>3 cm
Anomalies; hydrocolpos	Fewer	More (nearly all have urological defects)
Abdominal approach	Rarely	More likely
Bladder	Better spontaneous voiding	Clean intermittent catheterisation (CIC) far more often

5 · DSD RECONSTRUCTION — TIMING AND CONSENT

No trial compares early with late surgery.

- **For early surgery** gives pliable tissue; advocates seek a moratorium until consent.
- **Clitoroplasty** is only for severe hypertrophy (**Prader III–IV**) and is the most controversial component.

High-confluence vaginoplasty	Arguments
Early single-stage	Redundant clitoral/sinus tissue is usable; maternal oestrogen thickens tissue; a planned one-stage repair does better
After puberty	Infant repairs stenose; avoids dilation; allows consent; delay for the small vagina (**<3 cm**)
Hybrid	Clitorolabioplasty in infancy, vaginoplasty at puberty

> **KEY POINT**
>
> Without comparative data the decision is ethical, not evidential, and rests with parents and team. Endocrine Society CAH guideline (2018): families with experienced surgeons decide; the 2006 consensus limits surgery to Prader III–IV.

Body	Position
Lawson Wilkins and European paediatric endocrine societies' CAH group (LWPES/ESPE, 2002)	Surgery at 2–6 months; not between 12 months and adolescence
International intersex surgical society (ISHID) poll	Most prefer single-stage before 2 years
American Medical Association ethics council	Rejected a moratorium after 2 years' review

Body	Position
Advocacy groups; some US state bills	Moratorium until consent; bills not enacted

6 · OPERATIVE PRINCIPLES

Operate in this order: **clitoroplasty, then vaginoplasty, then labioplasty**.

Clitoral surgery	Technique	Verdict
Amputation; Lattimer recession	Excision; recession	Abandoned
Schmid (1961) MCQ	First corporeal excision preserving the neurovascular bundle	Basis of later techniques
Kogan (1983)	Subtunical excision, lateral Buck fascia incisions	—
Baskin (1999)	Defined neuroanatomy	**Ventral** incision spares the dorsal bundle
Pippi Salle (2007)	Corpora disassembled and buried in the labia	Reversible if male identity emerges

- The **Fortunoff (1964)** posterior perineal flap, now **omega-shaped**, underlies almost every repair. **Hendren and Crawford (1969)** showed that the **confluence level determines the type of vaginoplasty**.

Vaginoplasty	Indication	Principle	Pitfall
Cut-back	Labial fusion only	Incise sinus	Misses normal-calibre vagina
Flap	Low confluence	Posterior sinus opened, flap inset; confluence not moved	**Never for high confluence**: vaginal voiding, infection, incontinence
Pull-through	Very high	Vagina separated; sinus becomes urethra	If the vagina falls short, add flaps
Replacement	Absent vagina	See Ch 58	—

KEY POINT

Ask: **must the confluence be moved?** A flap brings skin to a low vagina; a pull-through takes a high vagina off the urethra. A flap on a high confluence is the classic error.

7 · LOW CONFLUENCE REPAIR

ALGORITHM

1. Ensure metabolic stability; give **stress-dose steroids in CAH** and antibiotics.
2. Do endoscopy; place a **Fogarty catheter in the vagina** and a Foley in the bladder.
3. Prepare the whole lower body for supine–prone rotation.

ALGORITHM

1. Place a glans traction suture.
2. Incise around the glans, keeping the inner prepuce for the hood (**second only to the glans in sensitivity**).
3. Raise the omega flap and make Y-incisions on the labia majora.
4. Deglove to the bifurcation, keeping the ventral plate intact.

- **Ventral subtunical excision**: incise Buck fascia on the **ventral-most** corpora and remove the erectile tissue, leaving the tunics and dorsal bundle untouched.

ALGORITHM

1. Flap vaginoplasty: raise the omega flap with its fat.
2. Free the posterior sinus off the rectum (**hardest step**).
3. Open the posterior sinus in the midline and **continue to normal-calibre vagina**.
4. Inset the flap.

Labioplasty: split clitoral skin forms the **labia minora**; the labia majora are moved down by **YV-plasty**.

> **KEY POINT**
>
> Two fixed points: never touch the dorsal neurovascular bundle; always open back to normal-calibre vagina.

8 · HIGH CONFLUENCE: PULL-THROUGH

- Only ≈ **5%** of CAH cases are high; they are commoner in pure sinus.
- The **critical step** is separating the anterior vaginal wall from urethra and bladder neck, which risks sphincter and nerves.
- **ASTRA** (anterior sagittal transanorectal approach) is done prone, dividing the anterior rectal wall: excellent exposure, no colostomy, continence unaffected.

ALGORITHM

1. Position prone; incise in the midline between sinus and rectum.
2. Divide the sinus up to normal vagina.
3. Free the anterior wall off the urethra **under direct vision**.
4. Close the sinus as the urethra in 2–3 layers; mobilise the vagina.
5. Turn supine for flaps and labioplasty.

Err on the side of the vagina; if it is short, add skin flaps.

9 · TOTAL AND PARTIAL UROGENITAL MOBILISATION

- **Total urogenital mobilisation (TUM, Peña 1997)** mobilises the whole sinus en bloc to the perineum without separation.
- **Never discard the sinus**: split **ventrally** it becomes the vestibule; split **dorsally** it becomes the anterior vaginal wall.

	TUM	Partial (PUM, Rink)
Extent	Retropubic and suprapubic	Stops at the **pubourethral ligament**
Use	Higher confluence; cloaca	Most sinus repairs
Risk	Sphincter/nerve injury, stress incontinence	Lower; convert to TUM if needed

10 · OUTCOMES OF SINUS/DSD SURGERY

- After infant vaginoplasty, **secondary introitoplasty** is needed after puberty in **25–100%**, because the orifice does not grow.
- **Azziz (1986)** MCQ reported poor coitus rates, but these reflect old **cut-backs**.
- **Confluence level, number of procedures and surgeon expertise** matter more than age; no data favour postpubertal surgery.
- **Elmasri meta-analysis (2018)**: female identity ≈ 89%; vaginal stenosis **27%**; only half had comfortable intercourse.

Two-procedure dilemma	Infancy	After puberty
Approach 1	Simple clitorolabioplasty	Extensive vaginoplasty

Two-procedure dilemma	Infancy	After puberty
Approach 2	Extensive single-stage clitorolabiovaginoplasty	Simple introitoplasty

Both usually mean two operations, so the key factor is a **surgeon or team with a special interest**.

11 · CLOACA — INITIAL MANAGEMENT

- **Hendren's rule**: never do a separate rectal pull-through; repair rectum, vagina and urethra in one stage.

ALGORITHM

1. Decompress the bowel.
2. Decompress the genitourinary tract.
3. Correct nephron-destructive anomalies.
4. Then do the definitive repair.

Colostomy	Advantage	Drawback
Low loop	—	Bowel hard to obtain for vaginoplasty; interferes with pull-through
Right transverse, divided (Hendren)	Left colic supply intact; colon available for vaginal replacement	Absorbs urine, causing acidosis
Descending, divided (Levitt, Peña)	Less resorbing mucosa; enough distal colon; less prolapse	Leave a long sigmoid

- Urine absorbed by colon causes **hyperchloraemic acidosis**. **Preserve the appendix** for a later Malone antegrade continence enema (MACE).
- **Urinary decompression**: **CIC is first line**, then a cloacal cut-back; vaginostomy or vesicostomy only if CIC fails.
- **Kidneys**: structural anomalies occur in **60–83%**; chronic kidney disease (CKD) in **75%** and end-stage renal disease in **17%**. Urodynamics are abnormal in **90%**.

12 · CLOACA — DEFINITIVE REPAIR

Repair at **6–12 months**, after urinary evaluation and exclusion of hydrocolpos.

Common channel	Urethra	Procedure (Wood algorithm)
<3 cm	**>1.5 cm**	**TUM**
>3 cm	or <1.5 cm	Urogenital separation
Any	Any	Never transabdominal TUM (ischaemia)

KEY POINT

Why 1.5 cm? A urethra >1.5 cm keeps the bladder neck above the urogenital diaphragm; mobilising a shorter one drags the bladder neck below the continence mechanism and causes leakage.

ALGORITHM

1. **PSARVUP (Peña; posterior sagittal anorectovaginourethroplasty)**: place catheters in each channel; position **prone jackknife**.
2. Mark the anal site with a muscle stimulator; incise in the midline.
3. Open the rectal fistula and mobilise the rectum.
4. Do TUM or vaginal separation.
5. Close the urethra in 2–3 layers with a **fat flap**; rebuild the perineal body; centre the rectum in the sphincter.

- It is **better to enter the vagina than the rectum or urethra**; separate it from the urethra by **sharp dissection, not cautery**.
- Aftercare: Foley for 2 weeks, rectal dilation from 2–3 weeks and colostomy closure at ≈ 3 months. Almost one-third empty poorly, so start CIC if in doubt.

13 · CLOACA — RESULTS AND FOLLOW-UP

- Outcome depends mostly on the **spine** and **common-channel length**.
- **Urinary**: about half void spontaneously. Peña: CIC in **28%** (channel <3 cm) vs **78%** (>3 cm). Review yearly for life.
- **Renal** function is preserved by aggressive bladder management. Faecal continence needs a bowel programme, then **MACE**.
- Vaginal stenosis or obstructed menstruation occurs in **36–41%**. Pregnancy is possible, with **caesarean preferred**.

ALGORITHM

1. Do US **6–9 months after thelarche**.
2. Repeat every 6–9 months until menses are normal.

65 Adolescent and Transitional Urology

1 · DEFINITIONS

- **Adolescent urology** bridges care for complex congenital conditions; the adult urologist lacks **whole-patient knowledge** (prior operations, altered anatomy, families).
- **Transition** (Got Transition) is the planned move from child/family-centred to adult/patient-centred care: **planning, transfer and integration**. Transfer is only one step.

> **KEY POINT**
>
> Renal function and HbA1c are expected to worsen, so **stability and retention in follow-up** are success.

2 · WHO NEEDS TRANSITION

Category	Examples	Implication
1. Complex congenital	Exstrophy, posterior urethral valves, complex hypospadias, spina bifida (SB)	Lifelong care
2. Treated, now quiescent	Vesicoureteric reflux	Early-adult review; may discharge with open return
3. Cured, with sequelae	Pelvic rhabdomyosarcoma	Late-effects surveillance
4. New adult concern	Fertility, pregnancy	Adult expertise

Failed transition (loss to follow-up, reliance on the emergency department) raises morbidity and mortality because **preventive surveillance is lost**.

3 · THE PROCESS

"Three Ps"	Content	Success measure
Preparation	Independence from 12–14 years; Transition Readiness Assessment Questionnaire (TRAQ)	All families involved

"Three Ps"	Content	Success measure
Plan	Follow-up schedule; self-care (catheterisation, irrigation, medications)	Shared with patient and future team
Provider of the future	May or may not be a new team	Named provider

- The **paediatric provider must initiate**, and never during a crisis; **pregnancy** is the exception and needs expedited transfer with co-management.
- Parents fear losing control, so give the patient **private time** (sexual function) and then ask what may be shared.
- Begin at about **12 years** (AAP) and complete by the late teens or early 20s; age **26** is the latest (US parental insurance ends). If independence is unachievable, plan custodianship.

Got Transition core element	Age (years)
Policy	12–14
Tracking, readiness, planning	14–18
Transfer of care	18–21
Transition complete (integration confirmed)	18–23

> **KEY POINT**
> Transfer is not the endpoint: transition ends only when integration into adult care is confirmed.

4 · ROUTES AND MODELS OF CARE

Patients arrive by transitioned referral (ideal), referral without transition, a new adolescent problem, or an **emergency after discharge**, which is the hardest and often the commonest.

Model	Strength	Drawback
1a Lifelong paediatric urologist	Continuity; suits limited life expectancy	Patient outlives the provider
1b Paediatric urologist in adult department	Continuity and adult expertise	Cost, privileges, geography
2 Transfer to an adolescent/adult specialist	Specialist coordinates the MDT	Cross-institution communication
3 Direct transfer, no plan	The default	Invisible until an emergency

Lifelong and multidisciplinary need are **independent**: exstrophy may need only a urologist, whereas SB needs the largest MDT. **Nursing** is the backbone.

5 · DEVELOPMENT AND CONSENT

- Avoid genital surgery at **3–6 years**; the 1975 AAP panel favoured genital surgery at **<6 months**.
- Augmentation waits until the child can catheterise and wants dryness.
- **Emerging adults (18–25 years)** consent for themselves unless they lack capacity.
- Patients with **SB** usually have normal intelligence but weak **working memory and executive function**.
- **Medical PTSD** causes avoidance and procedural panic; CBT may help.

6 · SETTING UP A SERVICE

- The core MDT includes paediatric and adult urologists, specialist nurses, a care coordinator, nephrologist and psychologist.

- Review consent **with the family, then with the patient alone**; standing orders (for example, urine cultures) avoid ED visits.

Barrier	Solution
Patient/parent: attachment, fear	Address the child directly early; readiness questionnaires
Paediatric provider: family trust	Ask "Am I the best person?"
Adult provider: **unfamiliarity with congenital conditions (biggest)**; finance	Partnerships, networks, training

120 Aging and Geriatric Urology

1 · CONCEPTS AND BIOLOGY

- **Chronological age is a poor descriptor**; physiological and functional age (function, frailty, geriatric syndromes) matter. The term "geriatrics" dates from Nascher (1909) MCQ .
- **Cellular senescence** drives ageing, either by telomere shortening (**Hayflick limit**, 1961) MCQ or by stress; its secretory phenotype (SASP) causes chronic inflammation.

2 · CLINICAL EVALUATION

- **Homeostenosis** is narrowed physiological reserve, so recovery after a stressor is poorer.

Tier	Content	Significance
Basic activities of daily living (ADLs)	Bathing, dressing, grooming, toileting, feeding, ambulation	**Bathing lost first**, feeding last; loss predicts dependency and postoperative outcome
Instrumental ADLs (IADLs)	Cooking, cleaning, transport, finances, telephone, medicines	Independent living

- A simple screen is **whether the patient can bathe independently**.
- Age is seldom an independent **surgical risk**; predictors are **longer operating time, poor ASA score and urgent surgery**. Optimise rather than "clear" the patient.

ALGORITHM

1. Screen for frailty before elective major surgery (SPAQI).
2. If positive, do a diagnostic frailty assessment.
3. Then do a comprehensive geriatric assessment (CGA).

- The **ACS NSQIP/AGS checklist** has 13 items, only 2 of which are organ systems; the geriatrician-led **CGA** adds physical performance, albumin, creatinine, BNP and FEV1.

3 · GERIATRIC SYNDROMES

- Geriatric syndromes feed one another, so treating one helps others. **Frailty** is decreased reserve causing vulnerability, distinct from comorbidity and disability.

	Accumulated deficits index	Fried phenotype
Method	Count failing systems	(1) weight loss >4.5 kg or 5%/year; (2) weak grip; (3) slow gait; (4) low activity; (5) exhaustion
Cut-off	Continuous	**≥3 frail; 1–2 pre-frail**
Validity	Does not predict postoperative outcome	Predicts death, dependence, falls, postoperative complications

Polypharmacy — Beers criteria

Harm often comes via the **prescribing cascade**, in which a side effect is treated with another drug.

Drug	Concern and action
Antimuscarinics (in dementia/delirium), tricyclics, first-generation antihistamines	Anticholinergic burden: delirium, falls, retention. Avoid combinations; no tricyclics in BPH
NSAIDs	GI bleeding; renal and heart failure at CrCl <30 mL/min
Systemic oestrogen; testosterone	Cancer, cardiovascular harm. **Vaginal oestrogen safe**; testosterone only for confirmed symptomatic hypogonadism
Ciprofloxacin, macrolides (**not azithromycin**), co-trimoxazole	Warfarin potentiation: monitor INR; dose-reduce at CrCl <30
Benzodiazepines	Cognition, gait, delirium. Avoid with opioids or ≥2 other CNS drugs
Nitrofurantoin	Listed at CrCl <30 or chronic use; usable in moderate impairment
Desmopressin; tramadol	Hyponatraemia: treat nocturia's cause; check sodium

KEY POINT

Four recurring mechanisms: anticholinergic burden, renal threshold (CrCl <30), warfarin potentiation, sodium disturbance.

Genitourinary syndrome of menopause (GSM)

- GSM causes dryness, dyspareunia and UTI **plus urgency and urgency incontinence**. Sex-steroid loss drives OAB in both sexes, including men on ADT or 5-ARIs.
- Treat with **vaginal oestrogen**. Prior breast cancer is not an absolute bar.
- A **urethral caruncle** causes painless spotting, so exclude gynaecological cancer.

4 · INCONTINENCE MANAGEMENT

- Incontinence is common but not normal; behavioural therapy comes first (see Voiding Dysfunction book, Ch 110). †Transient causes (DIAPPERS): delirium, infection, atrophic vaginitis, pharmaceuticals, psychological, excess output, restricted mobility, stool impaction.

P2 P3 *Overactive bladder and its drug therapy; OAB Pharmacotherapy — trospium quaternary, M3 selectivity, tolterodine QT, mirabegron BP/metoprolol, Beers avoid in dementia*

Not in the paper answer:

- Beers lists **oxybutynin and fesoterodine** as "avoid" in dementia, although the paper suggests fesoterodine for the elderly; trospium has the least central penetration.
- Fewer than half persist with **antimuscarinics** at 6 months; check the post-void residual after starting.

Nursing homes

- Catheter colonisation is **98%**, so treating asymptomatic bacteriuria (ASB) causes harm.

Loeb — catheter	Loeb — no catheter	NHSN
≥1 of: fever, new costovertebral tenderness, rigors, **acute mental status change**	Acute dysuria; or fever plus new urgency, frequency, suprapubic pain, haematuria or UI	Culture ≥10^5 CFU, no other source, and clinical signs

KEY POINT

A positive culture alone never qualifies; catheterised residents cannot report local symptoms, so systemic signs are used.

5 · UTI AND ASYMPTOMATIC BACTERIURIA

- **A causal link between confusion and UTI is not supported** (see Infections book, Ch 26).

- ASB affects up to 22% of older women and about 10% of older men and needs no antibiotics. USPSTF gives grade D against screening; IDSA advises treatment **only in pregnancy and before urological instrumentation**.

6 · CANCER, END OF LIFE

- **Prostate**: USPSTF grade D, so no PSA screening over 70. With life expectancy **<10 years**, plan non-curative care.
- **Bladder**: restaging TURBT for T1 is often omitted in the elderly yet improves survival. BCG is less effective (immunosenescence). Cystectomy at ≥75 gives comparable results with careful care.
- **Palliative care is not withdrawal**; it means aggressive symptom control after a goals-of-care discussion.
- The Age-Friendly "4 Ms" are Mobility, Medication, Mentation and what Matters; urologists must report **elder mistreatment**.

> **KEY POINT**
> Undertreatment by age (skipped restaging, withheld cystectomy) is as real a risk as overtreatment. Fitness and life expectancy decide.

7 · LANDMARK TRIALS

Trial	Design	Result	Take-home
ASPREE	RCT: aspirin vs placebo, healthy elderly	No disability-free survival gain; higher mortality	No primary-prevention aspirin
Testosterone Trials	RCTs, hypogonadal older men	No cognitive benefit; modest sexual-function gain	Not for cognition

Effective prevention includes vitamin D with calcium, antihypertensives, statins and exercise.

126 Management and Considerations in Transgender and Nonbinary Individuals

1 · DEFINITIONS AND TERMINOLOGY

- **Transgender and gender-diverse (TGD)** people have a gender identity that differs from their sex assigned at birth. Assigned sex, gender identity, gender expression and sexual orientation are **independent axes**.

Term	Meaning
AFAB / AMAB	Assigned female/male at birth
Transgender man / woman	Male identity, AFAB / female identity, AMAB
Transmasculine / transfeminine	Direction of gender, not a binary endpoint
Nonbinary	Neither male nor female
Gender dysphoria	Distress from identity–assigned sex discrepancy
Gender incongruence	Diagnostic term: marked, persistent incompatibility

Not all TGD people have dysphoria; some need only social transition, others gender-affirming hormone therapy (GAHT) and/or surgery.

2 · DIAGNOSIS AND ASSESSMENT

	DSM-5 (2013)	ICD-11 (2019)
Term	Gender dysphoria	**Gender incongruence** (replaced "gender identity disorder")
Criteria	Marked incongruence for ≥**6 months** , significant distress or impairment	Marked, persistent incongruence; desire for treatment of experienced gender
Chapter	Mental disorders	**Moved to Conditions Related to Sexual Health** MCQ

ICD-11 keeps a code for access but drops the mental-disorder label; **severe distress is not a prerequisite**.

WPATH Standards of Care 8 (SOC-8, 2022)

- Treat only marked, sustained incongruence, after excluding other causes and optimising coexisting conditions.
- Assess **capacity to consent**, and give **fertility counselling before any treatment**, not just surgery.
- **A single opinion** from a competent professional suffices for medical and surgical treatment.
- Before gonadectomy, **consider ≥6 months of hormones** unless not indicated.

> **KEY POINT**
> SOC-8 replaces gatekeeping with informed, capacity-based consent; capacity and fertility counselling stay firm because consequences are irreversible.

3 · GENDER-AFFIRMING HORMONE THERAPY (GAHT)

GAHT aims for **physiological** levels of the affirmed gender in patients with persistent dysphoria, capacity, majority age and controlled comorbidities.

Direction	Agent	Dose
Transfeminine — oestrogen	Oral oestradiol	2–6 mg/day
	Transdermal oestradiol	0.025–0.2 mg/day
	Oestradiol valerate/cypionate IM	5–30 mg 2-weekly or 2–10 mg weekly
Transfeminine — antiandrogen	Spironolactone	100–300 mg/day
	Cyproterone acetate	25–50 mg/day
	GnRH agonist	3.75 mg monthly / 11.25 mg 3-monthly
Transmasculine — testosterone	Enanthate/cypionate	100–200 mg 2-weekly
	Undecanoate	1000 mg 12-weekly (rare in the US: oil embolism)
	Gel 1.6% / patch	50–100 mg/day / 2.5–7.5 mg/day

> **KEY POINT**
> **Why the asymmetry?** Oestrogen alone does not suppress testosterone to the female range, so transfeminine therapy needs an antiandrogen too; testosterone suppresses the ovarian axis on its own.

Transfeminine

- **Ethinyl oestradiol is not recommended** (about 20-fold thrombosis). Use 17β-oestradiol, target **100–200 pg/mL**, and **use transdermal if VTE risk** (age >45 or previous VTE).
- **Spironolactone** is commonest in the US and **cyproterone** in Europe; GnRH agonists and bicalutamide are alternatives.

- **Irreversible effects are breast development and reduced fertility.** Risks include VTE, CVD, hypertriglyceridaemia, liver dysfunction and osteoporosis.
- Monitor levels 3-monthly in year 1, then 1–2 times a year, aiming for testosterone **<50 ng/dL**; check potassium and creatinine on spironolactone.
- **After gonadectomy, reduce oestrogen by 50–75%** (titrate to LH, FSH and bone density) and **stop the antiandrogen**.

Transmasculine

- Contraindications are pregnancy, unstable coronary disease and untreated polycythaemia; oestrogen-sensitive cancer needs oncology input. **Contraception is essential** because testosterone is teratogenic.
- Target testosterone **300–1000 ng/dL**; persistent menses respond to a progestin or endometrial ablation.
- **Irreversible effects are clitoral growth, facial hair, voice change and male-pattern baldness.**
- The main risk is **erythrocytosis**, so monitor testosterone and **haematocrit** 3-monthly in year 1, then 1–2 times a year.
- **After gonadectomy, the testosterone dose is not reduced**. **Nonbinary** regimens are individualised (for example, **nandrolone** for muscle with little hair).

4 · UROLOGICAL CONSIDERATIONS

Lower urinary tract symptoms have five contributors: **natal anatomy, hormones, surgery, pelvic floor dysfunction and psychosocial factors**.

- **Testosterone-induced vaginal atrophy** (dryness, bleeding, urgency) is treated with topical vaginal oestrogen.
- **After vaginoplasty**, about a third report altered voiding (**nocturia is commonest**); meatal stenosis, stricture and fistula also occur (see Ch 127).
- **After phalloplasty**, post-void dribble comes from pooling at the native urethra–pars fixa junction; stricture and fistula are far commoner (see Ch 128).
- Treat **pelvic floor** dysfunction before surgery; this eases neovaginal dilation.
- **Tucking** causes stasis, prostatitis-like symptoms, epididymo-orchitis and UTI. Exclude atrophy, tucking and pelvic floor causes before blaming the operation.

5 · FERTILITY PRESERVATION

Counsel **before hormones and/or surgery**, as WPATH, ACOG, ASRM and the Endocrine Society advise.

	Transfeminine	Transmasculine
Standard	**Sperm cryopreservation**; hold GAHT for ≥**3 months** before collection	**Oocyte or embryo cryopreservation** after ovarian stimulation
Prepubertal, or cannot pause hormones	Testicular tissue cryopreservation (experimental)	Ovarian tissue; no hormone pause needed
Obstacles	Ejaculation distress: vibratory stimulation, electroejaculation or surgical retrieval	Bleeding returns; transvaginal scans cause distress

Off testosterone, menses return in 3–6 months, and most conceive within 6 months.

6 · CANCER SCREENING — SCREEN THE ORGAN, NOT THE GENDER

- **Prostate (transfeminine)**: rare but may be aggressive. Offer shared decision-making on PSA at 55–69. **On GAHT, the upper limit is 1 ng/mL, or a rise of ≥0.3 ng/mL from nadir**, †because

androgen suppression lowers PSA. After vaginoplasty, feel the prostate through the **anterior neovaginal wall**.

- **Testis**: self-examination only.
- **Cervix (transmasculine)**: cytology 3-yearly at 21–29 years; at 30–65, cytology 3-yearly or hrHPV testing 5-yearly; stop after 65. Use a small speculum and **vaginal oestradiol for 1–2 weeks before**.

7 · AFFIRMING PRACTICE

- Keep a **detailed organ inventory** and use inclusive intake forms (name, pronouns).
- **Trauma-informed care** (SAMHSA) uses the "Four Rs": **Realise, Recognise, Respond, Resist retraumatisation**.

127 Male-to-Female Gender-Affirming Surgery: Vaginoplasty

1 · AIM AND PREPARATION

Many failures start with a patient who cannot dilate.

- **Mental health** input judges **surgical readiness**. Dysphoria-driven depression should **improve** after surgery, so it is no reason to defer; the key risk is **failure to dilate**.

Older criteria	WPATH Standards of Care 8 (SOC8)
Two assessments	**One** (inter-assessor agreement is high)
Mental health professional only	Any licensed clinician ≥ master's degree competent in transgender health
Hormones before gonadectomy	**6 months** suffices
Comorbidity "well managed"	Risks and benefits discussed
12 months' "real-life experience"	**Removed**: dangerous for those facing violence

Capacity is procedure-specific; if doubtful, a psychologist plans support.

> **KEY POINT**
> SOC8 replaces thresholds to clear with information to receive; capacity and fertility counselling stay mandatory.

- Stay within **1 hour's drive** of hospital for the first **3 weeks**.
- **BMI** mainly determines wound healing; no cut-off is proven, but many require **≤35 kg/m²**, and a higher BMI may suit vulvoplasty.
- **Permanent preoperative hair removal** of the scrotal–perineal graft is needed, because canal hair causes infection and hairballs.
- **Continue oestrogen perioperatively**; a 2022 meta-analysis found no added VTE risk.

Orchiectomy

- It may be the sole operation or a first stage, and afterwards spironolactone is stopped. Offer **fertility counselling and sperm banking to all**.
- Use a **vertical midline scrotal incision**; an inguinal incision may compromise the penile flap.

> **KEY POINT**
> Plan every orchiectomy as though vaginoplasty will follow: use a midline scrotal incision and leave cord fat for labial bulk.

2 · PENILE INVERSION VAGINOPLASTY (PIV)

P2 P4 *Penile inversion vaginoplasty technique — homologous tissue table, operative steps, dilation, complication rates*

Not in the paper answer:

- Pad the calves in lithotomy (**common peroneal nerve**) and give **tranexamic acid 1 g IV at the start**, as bleeding is front-loaded.

ALGORITHM

1. Foley and drain; mark adductor longus tendons (clitoral level) and dorsal midline (prevents torsion).
2. Harvest scrotal graft and small posterior rhomboid flap (relieves introital tension); defat, tubularise.
3. Orchiectomy, leaving cord fat; penile disassembly; corporal excision and clitoroplasty.
4. Urethroplasty, then canal dissection; line canal, labiaplasty, pack.

- **Disassembly**: the neurovascular bundle is best **left on the dorsal tunica**, and **all erectile tissue is excised**.
- **Urethroplasty**: excise bulbospongiosus **only after canal dissection**, as it covers a rectal injury, and spatulate downward for a seated stream.
- **Canal**: use repeated digital rectal examination, **preserve Denonvilliers fascia** and close it if breached. **Packing** ensures graft take.
- Keep the labia majora incision **≥1 cm** from the groin crease to avoid dehiscence.

Vulvoplasty (zero-depth)

Vulvoplasty suits those not wanting receptive intercourse and avoids **lifelong dilation**, but is not freely reversible. **Prior radical prostatectomy favours vulvoplasty**.

3 · ABDOMINOPERINEAL AND INTESTINAL OPTIONS

- **Peritoneal (Davydov-type)**: robotic dissection cut rectal injury from **~7%** to about 2.3%. The **rectum may mimic seminal vesicles**, and peritoneal defects must be **closed** to prevent hernia.
- Flaps are either **two-flap** (bladder and sigmoid peritoneum) or **pull-through** (abdominal wall peritoneum).
- **Intestinal** vaginoplasty suits severe penoscrotal hypoplasia or revision stenosis; IBD, adhesions, irradiation and short gut contraindicate it. **Dilation is still required.**

KEY POINT

Lining ladder: penile and scrotal skin first, then peritoneum, then bowel. Each adds depth but imports donor pathology; none removes the duty to dilate.

4 · COMPLICATIONS

Complication	Cause / key fact	Management
Wound dehiscence (commonest minor)	Introitus, dilator tension	Local care; granulation: curette or cautery plus steroid
Rectovaginal fistula (most devastating)	**0.8%** primary, 6.3% revision	Intra-op: watertight closure and bulbospongiosus flap, with or without diversion. Established: stop dilating and revise later, **or** delay closure
Urethrovaginal fistula (0.8–3.9%)	Missed injury, distal stenosis	Distal: often left. Bladder-neck (incontinence): transvaginal repair
Urethral stricture (1–6%)	Spraying, UTI, retention	Exclude pelvic floor dysfunction; dilate if short, else meatal spatulation

Complication	Cause / key fact	Management
Urethral bulge / upward meatus	Under-debulked spongiosum	Debulk, fix meatus downward
Vaginal stenosis	Poor dilation, scar, graft loss, pelvic floor dysfunction	Pelvic floor physiotherapy **first**; then scar excision and graft, or peritoneal/bowel augmentation
Prolapse (4–10%)	Poor graft take or fixation	Early repack; peritoneal fixation acts as a colpopexy

KEY POINT

Three failure modes: violated plane (fistulas), tissue not excised (bulge), graft not taken or maintained (dehiscence, stenosis, prolapse). Only the last depends on the patient, so dilation is part of the operation.

5 · OUTCOMES AND FOLLOW-UP

- **Median time to orgasm is about 180 days**. Depth is 12–16 cm, and self-lubrication is not the norm.
- Stay is about **5 days**; on **day 5 the pack and catheter come out** and dilation is taught.
- Follow-up is weekly for 3 weeks, then at 6 weeks, 3, 6 and 12 months, then annually. †The prostate is retained; survey it via the anterior neovaginal wall.

Period	Dilation (15–20 min each)
0–6 weeks	4 × daily
6 weeks–3 months	3 × daily
3–12 months	2 × daily
>1 year	Daily; less if stable

KEY POINT

The body treats the canal as a defect to close, so stenosis is a lifelong risk. Any narrowing means more dilation, contacting the surgeon and physiotherapy.

128 Special Urologic Considerations in Transmasculine Individuals

P4 *Enlist the steps in female to male gender reassignment surgery — staged sequence, vaginectomy, neourethra parts, metoidioplasty vs phalloplasty, scrotoplasty, prostheses, complication sites*

Not in the paper answer:

- Haemorrhage is the commonest vaginectomy complication, and a two-stage repair is standard for long strictures.

1 · SCOPE

- Goals are a male appearance, **voiding standing**, sensation and penetration (see Ch 126).
- **Early puberty blockade has a surgical price**: the vagina is harder to remove and small labia majora give less scrotal tissue.

2 · PREOPERATIVE CARE AND CONSENT

- **Urological screening is mandatory** (uroflow, PVR, renal ultrasound), and voiding dysfunction, stones and hydronephrosis are treated first.
- SOC7 requires ≥**12 months** of testosterone before genital surgery (†SOC8: 6 months) for **clitoral hypertrophy**, which plateaus at about 2 years.

Agent	Helps	Hinders
Testosterone	Clitoral hypertrophy aids metoidioplasty	Vaginal atrophy: harder vaginectomy
GnRH agonist	Prevents female characteristics	Small labia majora: less scrotoplasty tissue

Consent covers fistula, stricture, vaginectomy bleeding and organ injury, voiding dysfunction, UTI and implant problems.

3 · REMOVAL OF FEMALE STRUCTURES

- At hysterectomy the ureter lies beneath the uterine artery. †If the cervix is retained, screening continues.
- **Vaginectomy** risks haemorrhage, then bladder, then rectal injury. **Without vaginectomy there is no urethral lengthening** because of fistula risk.

ALGORITHM

1. Hydrodissect; circumferential introital incision.
2. Blunt rectovaginal plane; urethrovaginal plane **sharply, with no diathermy** (protects sphincter), then bladder plane.
3. If bladder injured: close, with or without cystoscopy.
4. Free levators, control paravaginal plexus; approximate levators.
5. Bulbospongiosus over urethra; layered closure.

Timing	Verdict
All-in-one	More major complications with phalloplasty; none extra with metoidioplasty
Early vaginectomy, phalloplasty later	Keeps blood loss away from free-flap perfusion
Two-stage (preferred)	Mastectomy, hysterectomy–oophorectomy; then vaginectomy and genital reconstruction

4 · GENITAL RECONSTRUCTION

- The fixed (perineal) and phallic neourethra both lack spongiosum. One **dorsal clitoral nerve** is coapted for sensation.
- **Bulbospongiosus over the fixed part** reduces fistulas. Scrotoplasty flaps run on the **external pudendal artery**.

	Metoidioplasty	Phalloplasty
Phallus	Own clitoris, **3–8 cm**	Flap
Erection	Spontaneous	Needs prosthesis
Sensation	Largely kept	Depends on nerve coaptation
Penetration	Usually not	Principal goal
Donor scar	None	Visible

5 · POSTOPERATIVE CARE AND CATHETERISATION

- The neourethral catheter comes out at **14–21 days**; the suprapubic catheter is clamped until voiding is normal.
- Catheterisation is hard because the curve into the native urethra is abrupt.

ALGORITHM

1. One gentle Coudé attempt, then one stylet-guided attempt.
2. If any difficulty, stop: flexible cystoscopy, guidewire, open-tip catheter.
3. In an emergency, place a **suprapubic catheter**.

6 · URETHRAL COMPLICATIONS

- **Fistula** is commonest at the phallic base, and small ones close with diversion. If one persists **>3 months**, excise and close in layers with fat interposition.
- **Stricture** is commonest at the **fixed-phallic anastomosis**. A plateau uroflow prompts urethrography; treat after **≥6 months**.

Option	Use
Dilation	Low-grade meatal stenosis
Internal urethrotomy	Low success; never repeat
Meatotomy / Heineke–Mikulicz	Short meatal / short non-obliterative
Excision and primary anastomosis	Short, near-obliterative anastomotic
Free graft	Needs flap support or dorsal inlay
Two-stage (standard)	Long/complex; retubularise ≥6 months later
Flap / perineal urethrostomy	Recurrent, or no further reconstruction wanted

- **Hair** is prevented with a hairless flap or laser.
- **Post-void dribbling** is near-universal; if severe, exclude sacculation or a vaginal-remnant diverticulum.

KEY POINT

The neourethra is skin in fat without spongiosum: fistulas and strictures cluster at poorly perfused flap ends, heal slowly and resist endoscopic cure.

7 · PROSTHESES

- Insert an erectile prosthesis ≥**1 year** after phalloplasty, once there is **protective sensation** and a stable urethra.
- Infection or erosion needs urgent explantation, with reimplantation ≥6 months later.

★ Putting it together

A · NEWBORN WITH ATYPICAL GENITALIA

ALGORITHM

1. **Emergency is adrenal:** CAH is commonest (half of all; 90% of 46,XX). Salt wasting (low Na, high K, shock) appears at day 5–7 (see Ch 59).
2. **Tests:** karyotype (SRY), 17-hydroxyprogesterone after 36 h, electrolytes, pelvic ultrasound.
3. **Team:** †defer sex assignment until MDT evaluation.
4. **Gonads:** dysgenetic with Y (30–45% tumour risk): prophylactic gonadectomy.

B · UROGENITAL SINUS AND CLOACA

KEY POINT

True urethral length outweighs common-channel length. Low confluence needs a flap; high needs pull-through or urogenital mobilisation. Clitoroplasty only for Prader III–IV, sparing the dorsal bundle. Cloaca: colostomy, decompress hydrocolpos, imaging with endoscopy, then posterior sagittal repair; seek VACTERL and tethered cord (see Ch 60).

C · ABSENT VAGINA

Cause	Key feature	Management
Mayer–Rokitansky–Küster–Hauser	46,XX, no uterus; †image kidneys	Dilation first; else McIndoe, bowel, buccal or peritoneal
Complete androgen insensitivity	46,XY, testes, no uterus	Dilation usually suffices
Septum / imperforate hymen	Uterus present, haematocolpos	Hymen: incision, not aspiration; septum: excision

D · IMMUNOTHERAPY AND HEREDITARY CANCER

ALGORITHM

1. **Checkpoints:** PD-L1 silences PD-1-positive T cells; CTLA-4 acts at priming. †Immune adverse events grade ≥2: hold drug, give steroids; endocrinopathies need lifelong replacement (see Ch 24).
2. **Hereditary:** VHL (clear cell; belzutifan), HLRCC (aggressive papillary), Birt–Hogg–Dubé (hybrid oncocytic), Lynch (upper-tract urothelial).
3. BRCA prostate cancer: PARP inhibitor. FGFR-altered urothelial cancer: erdafitinib (see Ch 25).

E · OLDER PATIENT BEFORE SURGERY

KEY POINT

Function, not age, predicts outcome. Screen frailty, then do a comprehensive geriatric assessment (cognition, delirium, nutrition, falls, polypharmacy). Avoid anticholinergics (prefer a β3-agonist). For new incontinence, seek transient causes first (†DIAPPERS) (see Ch 120).

F · TRANSGENDER AND NONBINARY PATIENTS

ALGORITHM

1. Screen the organs present: retained prostate (†PSA lower on oestrogen) and retained cervix (see Ch 126).
2. Vaginoplasty: stenosis and rectovaginal fistula (most devastating) (see Ch 127).
3. Phalloplasty: fistula and stricture, worse without vaginectomy (see Ch 128).